

**New trading
relationships can
change the
dominant vehicle.**



**Centre for Blockchain
Technologies**

The Making of Dominant Vehicle Currencies

Evidence from DeFi
Jiahua Xu

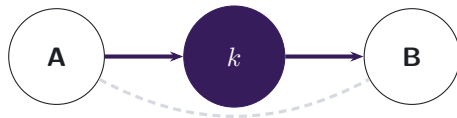
A cross-border payment needs someone to make the FX market

THE LIQUIDITY-PROVIDER PROBLEM

A payment provider receives one currency but must deliver another it does not hold.

more than 85%

of FX transactions involve the U.S. dollar



Currency k bridges an illiquid pair.

vehicle currency

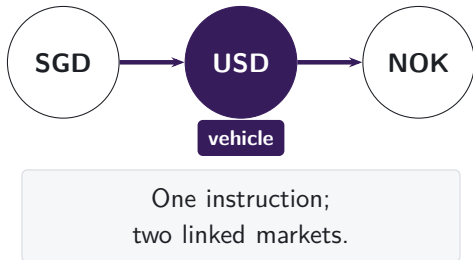
Winning the middle concentrates trading and liquidity in the vehicle.

Pool routes reveal the vehicle currency

SINGAPORE PAYMENT ANALOGY

A Singapore-dollar payment to Norway may be converted through US dollars.

Conventional turnover data record two legs.
Pool logs preserve their connected route.



DeFi is on-chain finance executed by smart contracts. Decentralised exchanges trade tokens through liquidity pools; **stablecoins** target a reference currency, usually the U.S. dollar.

One transaction reveals the connected route

ROUTE PANEL

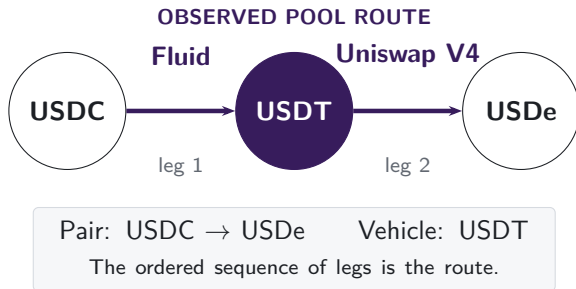
475 million

pool-level swaps

**November
2018–June
2026**

9

DEX deployments



V2 turns a mandate into a market choice

V1: NOVEMBER 2018



217,003

of token-to-token routes
pass through ETH

V2: MAY 2020

95.5%

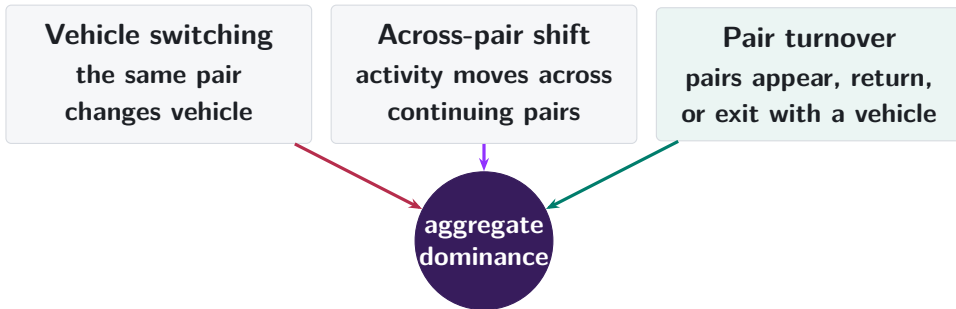
of single-leg trades use a WETH pool
in 2026

97.9%

of new token combinations include
WETH

The rule disappears; inherited liquidity can remain.

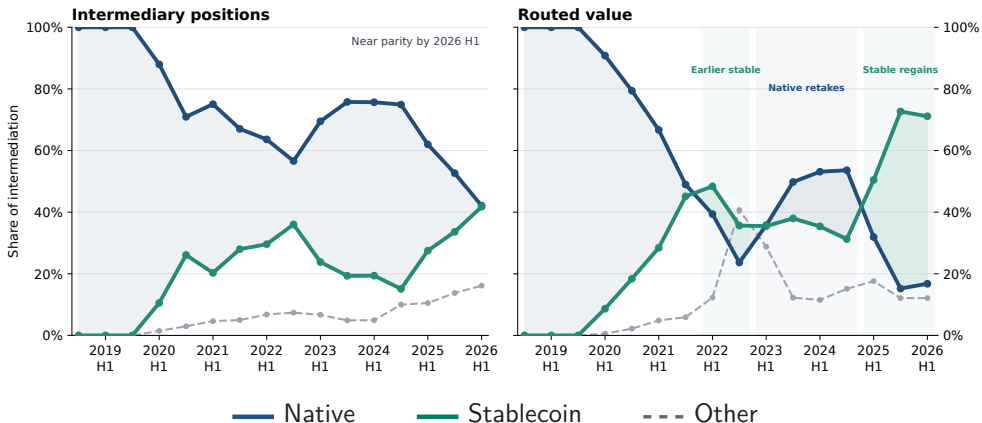
Aggregate dominance can change in three ways



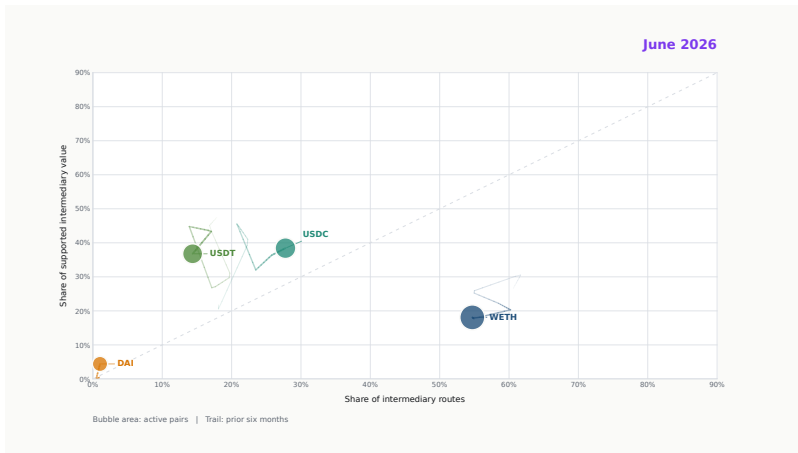
The decomposition locates the rotation; later evidence explains persistence.

Stablecoins regain the routed-value lead by 2026 H1

ALL ROUTE LENGTHS



Vehicle leadership turns over through time



► Play 18-second film

Pair formation supplies most of the rotation

EXACT DECOMPOSITION IN PERCENTAGE POINTS (PP): 2024 H1 TO 2026 H1

$$\Delta S = \underbrace{\bar{W} \sum_p \bar{\omega}_p \Delta s_p}_{\text{switching}} + \underbrace{\bar{W} \sum_p \bar{s}_p \Delta \omega_p}_{\text{reallocation}} + \underbrace{(\bar{S}_C - \bar{S}_E) \Delta W}_{\text{continuing weight}} + \underbrace{(1 - \bar{W}) \Delta S_E}_{\text{period-specific pairs}}$$

-0.1 pp +8.4 pp -0.5 pp +17.8 pp

STABLECOIN SHARE **16.9%** → **42.4%**

First-observed pair entry supplies +20.1 pp of +25.5 pp; switching gives -0.1 pp.

A pair carries its first vehicle into later trading

STABLECOIN SHARE AFTER ENTRY, PER 10 PP MORE STABLECOIN USE ON THE ENTRY DAY

	Days 1–30	Days 31–120
All retrading pairs	+8.92 pp (0.15 pp)	+8.40 pp (0.18 pp)
At least 5 entry routes	+9.70 pp (0.19 pp)	+9.31 pp (0.44 pp)
At least 10 entry routes	+9.65 pp (0.32 pp)	+9.35 pp (0.60 pp)

Near one-for-one persistence, however precisely entry is measured.

Two-leg capital opens the stablecoin contest

THE EVENT DATE USES PRIOR-DAY CAPITAL ONLY

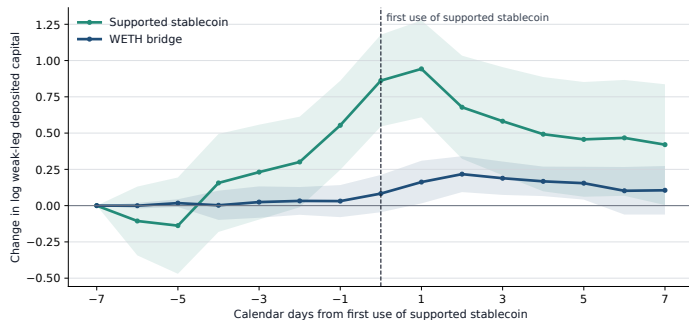


Among first-month adopters, **62.4%** use a supported stablecoin again in days 30–119;
stablecoins carry **8.2%** of their routes in that window.

Viability comes first. Dominance develops more slowly.

Capital arrives before the first stablecoin route

SUPPORTED STABLECOIN FIRST USED AT DAY ZERO



BEFORE FIRST USE

+0.86

stablecoin bridge, day -7
to 0

AFTER FIRST USE

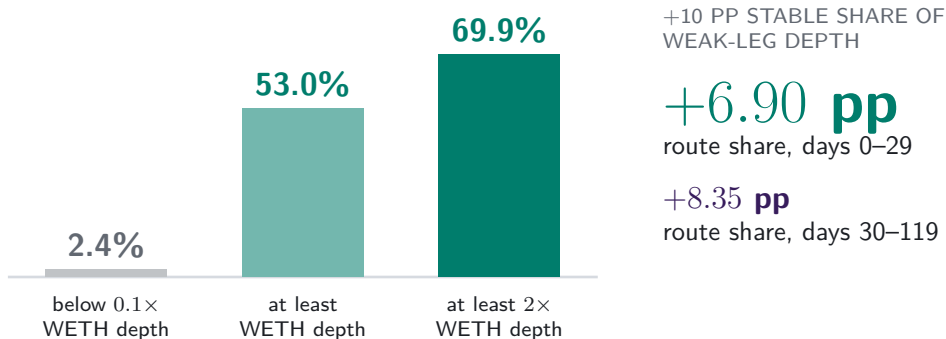
-0.44

stablecoin bridge, day 0
to +7

Capital builds into first use, then partly unwinds.

Relative depth divides trading after the bridge forms

STABLECOIN ROUTE SHARE IN THE FIRST 30 DAYS AFTER PERSISTENT SUPPORT



Vehicle competition turns on relative depth, leg by leg.

Current prices can overturn incumbency

STABLECOIN AND WETH PATHS BOTH FEASIBLE: SAME PAIR, INPUT, PRETRADE STATE

ENTRY FAMILY
RETAINED

83.1%

of routes at the first
feasible contest

INCUMBENT
RETURNS MORE
OUTPUT

93.3%

retain the incumbent

CHALLENGER
RETURNS MORE
OUTPUT

27.2%

retain the incumbent

Pair and date fixed effects: challenger price leadership changes incumbent retention by
-58.08 pp.

Incumbency holds until the price lead moves.

When the price lead flips, route share follows

WITHIN-PAIR EXACT-PRICE CROSSING



ADDITIONAL FALL AT THE PRICE CROSSING

-29.0 pp

CHALLENGER PRIOR WEAK-LEG CAPITAL SHARE (+10 PP)

3.70 pp

probability the price lead lasts one month

Prices move flow. Earlier depth predicts which lead lasts.

Shortfalls cluster in young pairs and small trades

EXACT STABLECOIN-VERSUS-WETH COMPARISON; INPUT-VALUE WEIGHTS

ALL INPUT VALUES: GROSS → NET OF GAS

7.46 → 7.65 bp

nearly unchanged

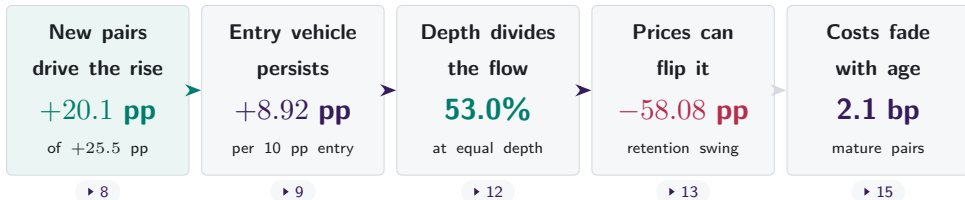
INPUTS \$100–999: GROSS → NET OF GAS

16.02 → 22.01 bp

fixed toll matters



The making of a dominant vehicle currency



Two margins of currency competition

Which vehicle a new pair builds around; which survives later price and depth changes.

Dominance is decided where new markets form

Payment rails Nexus, Rialto, mBridge: access rules pick the currency that new corridors inherit. ▶ 8 ▶ 9

Stablecoin oversight Most routed value now crosses a stablecoin in the middle; stress there reaches every pair routed through it. ▶ 6

Challenger currencies Cheaper quotes in existing corridors dethrone nobody; challengers win by being liquid at formation. ▶ 8 ▶ 12

Liquidity at formation beats price competition later.

What we are asking next

- Who supplies the bridge? Provider networks across pairs; issuer circulation into bridge capital; fee and incentive events.
- Does settlement architecture move liquidity? Uniswap V4 flash accounting against V3.
- Who earns the rents of dominance: traders, providers, or issuers?
- The same entry-margin account, taken back to conventional FX.

Comments on any of these are very welcome.

Appendix map

Definitions

- A1 Routes
- A2 Assets and pegs
- A3 Samples

Validation

- A4 Pretrade state
- A5 Pool pricing
- A6 Uniswap v1

Robustness

- A7 Timing and venue
- A8 Weights
- A9 Endpoints

Markets

- A10 Liquidity quantities
- A11 Coordination
- A12–A13 References

Further results

- A14 Network position
- A15 Endpoint demand
- A16–A17b Composition
- A18–A23 Depth and prices
- A24 Persistence robustness
- A25 LP risk

Pool data may start after the user instruction

USER INSTRUCTION RECORDED BY THE EXPLORER

Transaction details # 0x

⚡ Swap 460.00K PYUSD for 460.10K USDe on 0x

THE DECISIVE SETTLEMENT TRANSFERS

 USDC

ERC-20 Transfer

Market Maker: 0x51c7..... → Mainnetsettler

 USDC

ERC-20 Transfer

Mainnetsettler → Fluid: Liquidity

The maker supplies the USDC used by the Fluid leg. The instruction is PYUSD → USDe.

A1. One reconstructed route is one unit



one route unit
two swap legs

- Each linked $i \rightarrow k \rightarrow o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k .

How we reconstruct a route from one transaction

Direct swap

one transaction



Sequential vehicle route

one transaction



Parallel split

one transaction



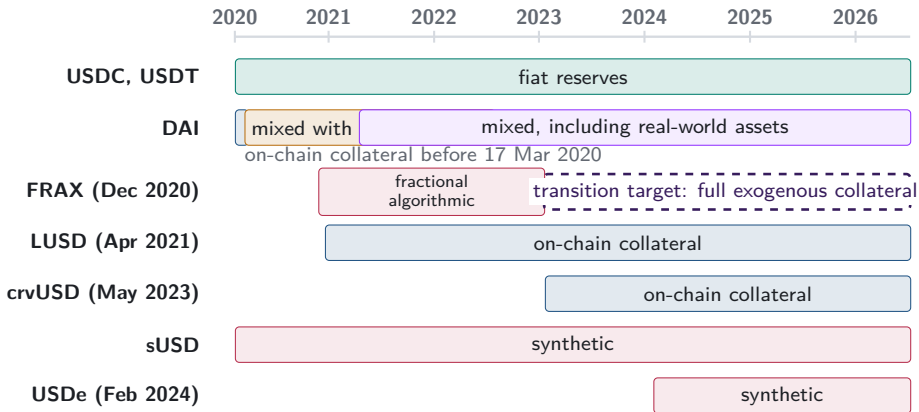
Round trip

one transaction



Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level routes reveal that choice inside aggregate turnover.

A2. Stablecoin backing changes over time



Peg parity preserves currency identity

Bahamian dollar $\longleftrightarrow$ **U.S. dollar**

one-for-one peg

Separate currency, monetary arrangements,
and exchange controls

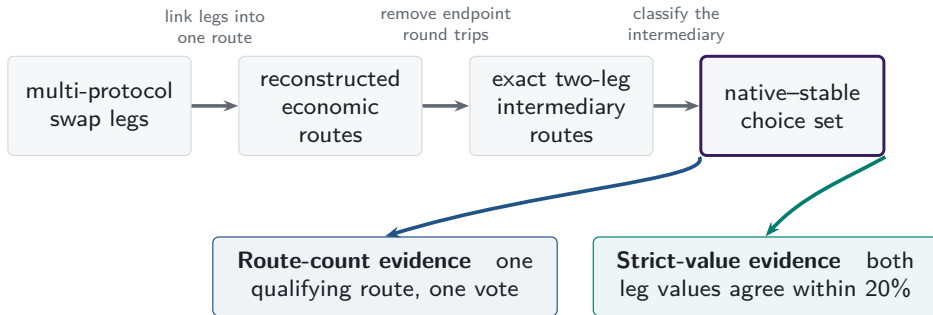
USDC $\longleftrightarrow$ **USDT** $\longleftrightarrow$ **DAI**

dollar target

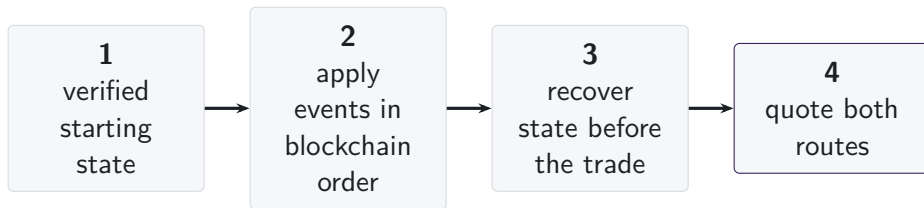
Separate issuers, redemption, peg risk, and
pool liquidity

Lower divergence loss may attract liquidity providers; issuers or affiliated market makers may also seed conversion pools. Peg stress reverses the comparison.

A3. One route universe supports two measurements



A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hook-dependent pricing
Constant product	■	—	—	—	—	■	—
Concentrated liquidity	■	■	—	—	—	■	—
StableSwap families	■	—	■	—	■	■	—
Weighted pools	■	—	—	■	■	■	—
V4 singleton	■	■	—	—	—	■	□

Executable-quote validation

■ recovered pricing input

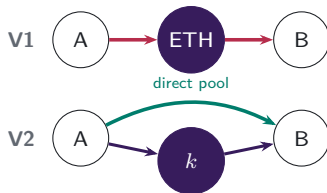
□ hook-dependent pricing unavailable

Protocol mechanisms studied

V4 singleton flash accounting and netting remain in scope; A5.3 shows the settlement mechanism.

A5.1. Pool formation determines available paths

V1 PROTOCOL RULE $\rightarrow$ V2 MARKET CHOICE



V1 admits only ETH-token pools. V2 permits any ERC-20 pool, so market creation determines whether direct and vehicle paths coexist.

Which paths exist?

Economic implication: pool creation changes the feasible route set.

A5.2. Executable depth is path-specific

V3 CONCENTRATED-LIQUIDITY POSITIONS



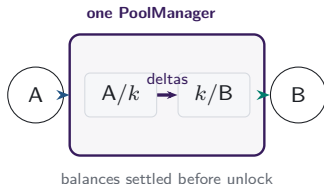
LPs choose a price range and fee tier for each pool. Positions covering the current price determine active liquidity in the direct pool and each vehicle spoke.

Which path offers depth at this trade size?

Economic implication: executable depth is path- and size-specific.

A5.3. Shared accounting moves route settlement

V4 SINGLETON AND LOCK ACCOUNTING



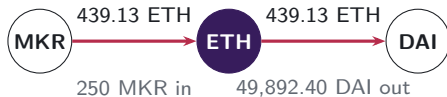
One PoolManager holds all pools. During a lock, the two route legs update balance deltas; the caller settles its obligations before the lock ends.

Where is the route settled?

Economic implication: shared accounting changes where route obligations settle.

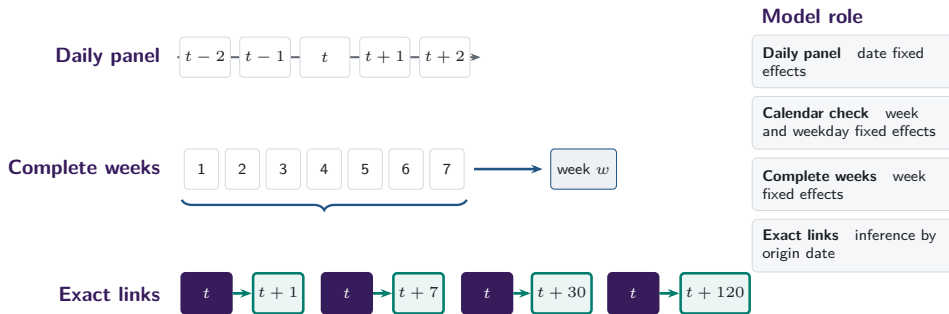
A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.



same transaction, matching ETH flow

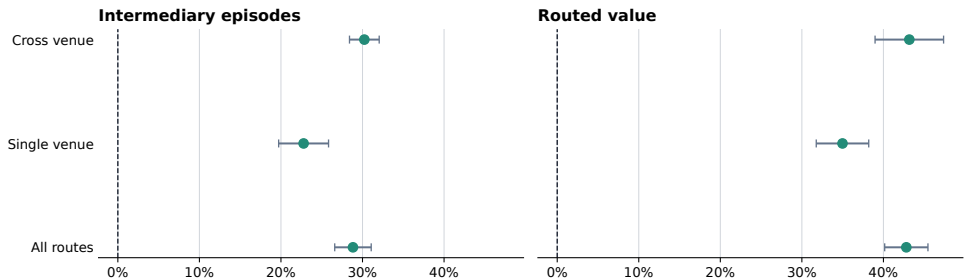
A7. Daily and weekly frequencies answer different questions



Exact horizons use the required target calendar date.

A7b. Stablecoin share rises within each venue scope

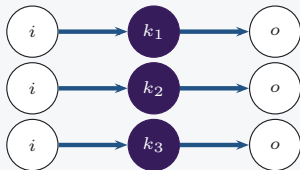
Stable share rises within realised route scopes



Points are 2024-to-2026 changes; bars are 95% HAC intervals.

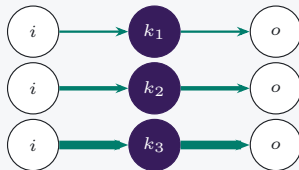
A8. Count versus value weighting

Count weighting



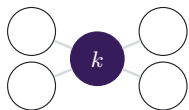
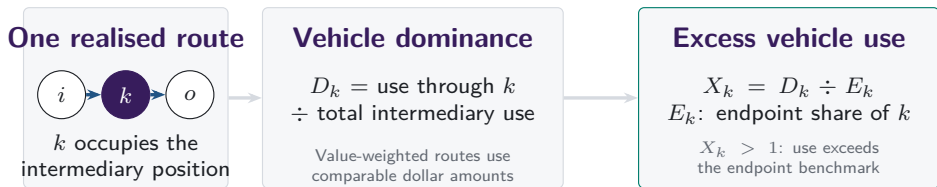
one qualifying route = one unit

Value weighting

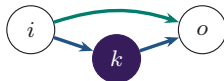


supported dollar value sets the weight

Vehicle dominance aggregates realised intermediary choices



Network position



Execution cost

Endpoint use is the benchmark; network position and same-state cost remain separate.

A9. Intermediary and route-endpoint use are separate

Intermediary share

$$I_{k,t}^N = \frac{n_{k,t}^I}{\sum_j n_{j,t}^I}$$

How often asset k sits inside a route.

Endpoint share

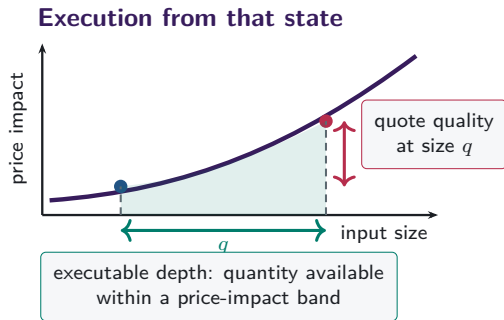
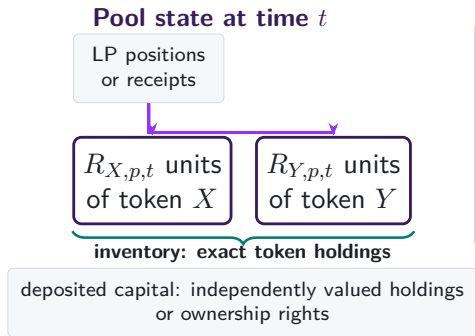
$$E_{k,t}^N = \frac{n_{k,t}^E}{\sum_j n_{j,t}^E}$$

How often k appears at either observed route endpoint.

Observed pool-route endpoints supply the benchmark.

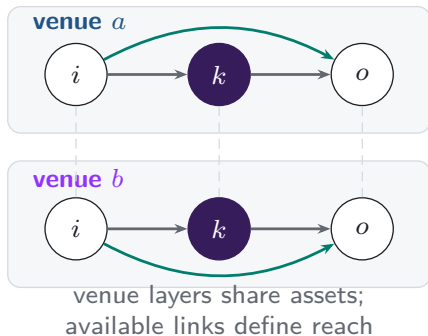
Excess use compares intermediation with observed route-endpoint use on the same support.

A10. Capital, inventory, and depth differ



A11. Additional venues expand the feasible route set

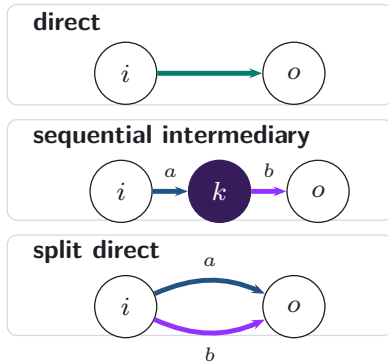
Feasible venue layers



realised
execution

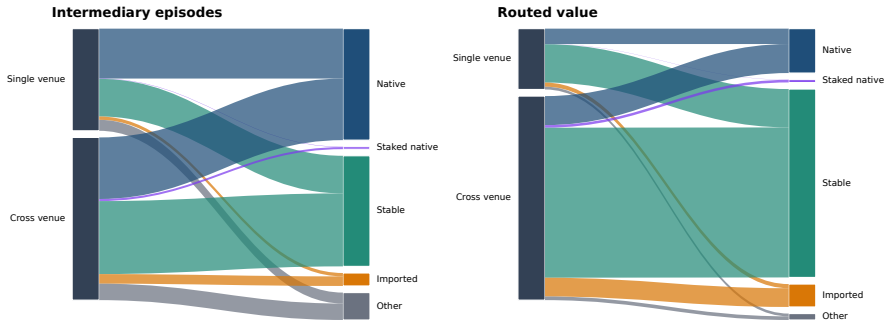


Realised route topology



A11b. Venue scope and vehicle type in 2026

Integration and vehicle composition are separate margins in 2026



Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Adams et al. (2024), *Uniswap v4 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Rinaldo, decentralised exchange costs and market structure
- Lehar and Parlour (2024), *Journal of Finance*
- Millionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
- Xu, Paruch, Cousaert and Feng, AMM design and slippage

A14. WETH stays the graph hub as realised use shifts

ALL ROUTE LENGTHS; MONTHLY TRADING GRAPHS

WETH BETWEENNESS

#1

in all 8 annual graphs

0.925

in 2026 H1

WETH REALISED SHARE

76.2%

2024

42.4%

2026 H1

USDC + USDT REALISED
SHARE

14.9%

2024

37.2%

2026 H1

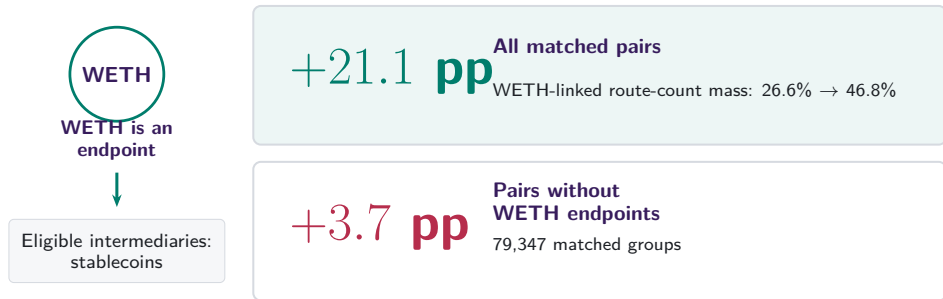
Binary network position changes much less than actual intermediary use.

A15. Endpoint demand predicts intermediary use

	Intermediary share ($I_{a,t}$) [pp]			
	(1) Pooled	(2) Date FE	(3) + demand	(4) + currency FE
Native currency	+34.56 (23.04)	+34.55 (23.04)	+0.05 (0.56)	—
Stablecoin	+2.42* (1.32)	+2.42* (1.32)	+0.85* (0.50)	—
Endpoint-demand share ($D_{a,t}$)	—	—	+1.59*** (0.03)	+0.98*** (0.30)
Date fixed effects	No	Yes	Yes	Yes
Currency fixed effects	No	No	No	Yes

After date and currency effects, endpoint demand passes through almost one for one into intermediary use.

A16. Pair composition contains a mechanical component



Changes within fixed pairs are +0.2 pp overall and +0.3 pp after removing WETH endpoints.

A17. Endpoint direction splits count and value

CONTRIBUTION TO THE STABLECOIN-SHARE INCREASE

Route count: +25.4 pp

Other pairs  +15.2 pp

One native, one stable  +9.2 pp

Two stable endpoints  +1.0 pp

Routed value: +43.9 pp

Other pairs  +20.6 pp

One native, one stable  +10.1 pp

Two stable endpoints  +13.2 pp

USDT supplies +13.7 pp of the +13.2 pp two-stable-endpoint value channel.

A17b. Rotation survives time and endpoint restrictions

**-0.4 pp to
+1.0 pp**
within-pair term across
all adjacent H1
comparisons

**1.1% →
9.0%**
route-count share with
nonvehicle endpoints

**0.9% →
39.1%**
supported-value share with
nonvehicle endpoints

Pair entry and exit contribute +7.2 pp by count and +33.8 pp by value.

A18. Local depth remains informative alongside network reach

84.1%

of routes use the deepest supported bridge

+6.79 pp

route share per log point of weak-leg capital

+9.10 pp

route share per log point of same-day network reach

Local two-leg depth and broader network position both matter.

A19. First-use capital lies in active pools

SUPPORTED STABLECOIN, DAY -7 TO FIRST USE

92.5%

of first-use capital lies in
continuing pools

+0.16

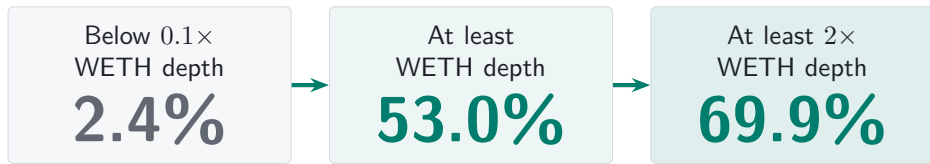
continuing-pool growth
relative to WETH

+6.9 pp

greater probability of a
newly active pool

A20. Relative depth predicts route flow

FIRST 30 DAYS AFTER PERSISTENT STABLECOIN SUPPORT



Vehicle competition turns on relative depth.

A21. Usable depth turns support into first use

PROBABILITY OF FIRST STABLECOIN USE WITHIN 30 DAYS

BELOW $0.1\times$ WETH DEPTH

42.6%



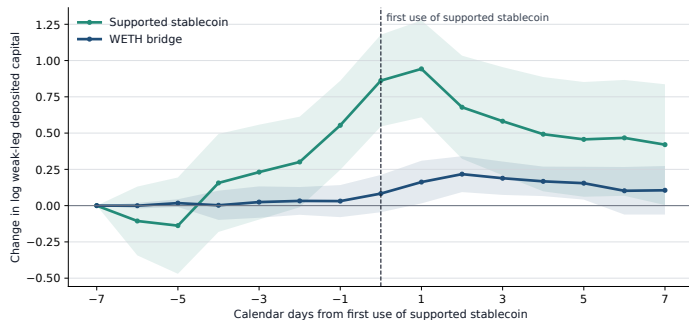
AT LEAST $0.1\times$ WETH
DEPTH

84.1%

Positive pool capital becomes useful as both route legs deepen.

A22. Bridge capital builds before first use

SUPPORTED STABLECOIN FIRST USED AT DAY ZERO



BEFORE FIRST USE

+0.86

stablecoin bridge, day -7
to 0

AFTER FIRST USE

-0.44

stablecoin bridge, day 0
to +7

Capital builds into first use, then partly unwinds.

A23. Broader price search mostly changes the venue

SHARE OF ROUTES WITH A QUOTE MORE THAN 1 BP BETTER THAN EXECUTION

SAME VEHICLE,
USED VENUES

6.6%



SAME VEHICLE, ALL
EXACT VENUES

44.5%



ANY VEHICLE OR
DIRECT

46.4%

Opening other vehicles adds 2.0 pp; stablecoin share changes by -1.2 pp.

A24. Persistence is equally strong on busy entry days

EFFECT OF 10 PP MORE STABLECOIN USE ON THE ENTRY DAY

	Days 1–30	Days 31–120
At least 5 entry routes	+9.70 pp	+9.31 pp
At least 10 entry routes	+9.65 pp	+9.35 pp

Stronger entry measurement leaves the persistence estimate intact.

A25. Divergence risk helps locate bridge capital

EFFECT OF 10 PP MORE PRIOR RELATIVE VOLATILITY

LOG BRIDGE CAPITAL

—0.117

current date (0.049)

—0.093

30 days later (0.029)

MEDIAN RELATIVE
VOLATILITY

Native 126.9%

Stablecoin 148.4%

Stablecoin bridge has lower risk in
29.4% of pair-months.

Risk shapes local capital; the rotation extends beyond it.